Decision-Model-Guided Function Calling with Cumulative Verification Feedback

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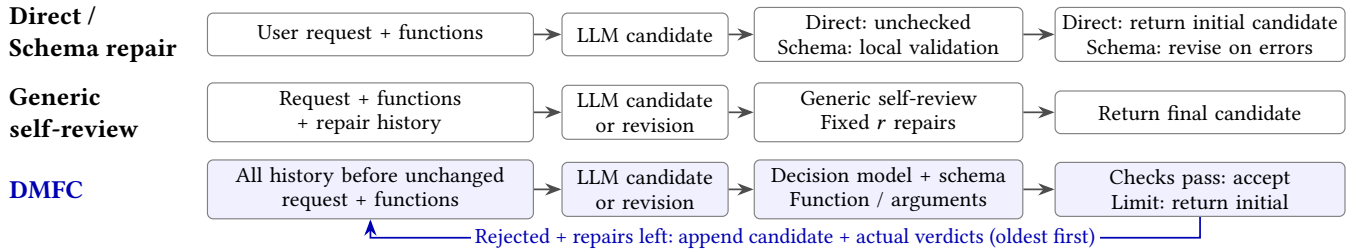


Figure 1: Comparison of evaluated mechanisms (schematic; not measured results). DMFC retains all rejected candidates and actual verdicts before the unchanged request, checks arguments only for a call, and executes no external function during repair.

Abstract

Function calling requires choosing a function and arguments, or withholding a call. We present **Decision-Model-Guided Function Calling (DMFC)**, which combines schema validation with binary function and argument judgments. Rejected candidates and actual feedback accumulate before the unchanged request; repair never executes functions. With one decision model and at most three repairs, three-run BFCL accuracy rises from 64.1% to 74.8% with GPT-5-nano and 81.1% to 82.1% with GPT-5.4-mini, but falls from 88.3% to 87.0% with Claude. On an exploratory 156-case API-Bank first-call adaptation, accuracy rises from 60.0% to 66.5%, 78.2% to 79.5%, and 86.3% to 87.2%, respectively. A single-run BFCL ablation finds a 1.1-point advantage over latest-only feedback. Effects vary across generators and require additional inference time and cost. API-Bank's input was refined after exploratory outcomes; this evaluation measures call prediction, not execution success.

Keywords

function calling, decision models, verification, in-context learning

1 Introduction

Function calling maps a request to a function and arguments, or to no call when the available functions are irrelevant. Errors can produce unintended actions: a schema-valid call may select the wrong API, reverse parameter roles, or act unnecessarily. BFCL evaluates names, arguments, and call relevance [8].

Tool-use training improves API selection and generation [9, 10]. Inference-time approaches include self-generated feedback [6], tool/parameter verification [7], tool-error critique [2], and reasoning with execution observations [16]. Decision models supply targeted judgments: TypeSafe's *Jev* exposes binary checks [13], while JEV-as-a-Judge studies acceptance and escalation for completed outputs [4]. These precedents motivate testing cumulative feedback before execution.

Three questions guide the study. **First**, do semantic checks improve on structural repair? Our schema baseline raises call accuracy only from 63.8% to 65.8%. **Second**, do separate function and argument judgments help? Conditional argument scores omit wrong-function and no-call decisions, so we measure full-call correctness and component ablations. **Third**, does retaining earlier rejections help within the same budget? We compare full history with latest-only feedback while measuring time and cost.

Decision-Model-Guided Function Calling (DMFC) checks function choice and applicable arguments, plus local schemas. Rejected candidates and actual verdicts accumulate before the unchanged request, oldest first; the decision model sees only the current candidate. Only a candidate passing all applicable checks replaces the initial answer. Repair never executes external functions (Figure 1).

We compare two OpenAI generators and Claude with one decision model on BFCL and an API-Bank first-call adaptation, sharing initial candidates in three runs. DMFC improves BFCL accuracy by 10.7 points with nano and 1.0 with mini, but reduces Claude accuracy by 1.3 points. API-Bank gains are 6.4, 1.3, and 0.9 points, respectively; a BFCL ablation finds a 1.1-point benefit from full history. Our contributions are the separate-check cumulative repair design and a paired assessment of accuracy, ablations, and latency–cost trade-offs. API-Bank's input instruction was refined after exploratory outcomes, so this is limited transfer evidence.

2 Methodology

2.1 Task and Prior Approaches

Let x be a request, $\mathcal{F}$ the available functions, and $c = (f, a)$ a proposed function and arguments, or $c = \varnothing$ for no call. We study a single call-or-no-call decision before execution. Correctly withholding an irrelevant call matters because valid syntax alone cannot prevent an unintended external action. We evaluate one next call,

not interactive tool execution; API-Bank inputs may contain several prior dialogue turns.

Toolformer learns when to use APIs [11]. τ-bench evaluates tool-agent interactions under user and policy constraints [15], while API-Bank includes executable API calls in dialogue [3]. Our task isolates a single proposed call before execution. Many-shot learning studies larger demonstration sets [1]; our history instead contains failed attempts at one request.

2.2 Decision-Model Verification

DMFC separates a generator G from a decision model J:

$$c_t = G(H_t, x, \mathcal{F}), \qquad p_f^{(t)} = J_f(x, \mathcal{F}, c_t), \quad p_a^{(t)} = J_a(x, \mathcal{F}, c_t). \tag{1}$$

Here H_t is the rejected-call history, initially empty. The function question checks whether the selected function, including no call, fits x. The argument question checks required values, support in x, and parameter roles; it is skipped for no call. A readable call with missing arguments still receives both checks. Passing the argument check alone cannot establish correct function selection.

A decision model supplies external binary judgments; our evaluated implementation uses TypeSafe's `jev-1.13.0` [13]. Its `Noul` primitive returns the probability of *yes*, not a separate confidence score [12, 14]. JEV-as-a-Judge studies confidence-based acceptance and escalation [4]; process supervision trains step verifiers for mathematical reasoning [5]. These are adjacent uses of verification, not evidence that other decision models have been tested in DMFC.

2.3 Acceptance and Cumulative Repair

Local validation checks readability, function availability, required arguments, and types; $S(c) = 1$ denotes success. All applicable checks must pass on the same candidate:

$$A_t = S(c_t)\mathbf{1}[p_f^{(t)} \geq \tau_f] \begin{cases} 1, & c_t = \varnothing, \\ \mathbf{1}[p_a^{(t)} \geq \tau_a], & \text{otherwise.} \end{cases} \tag{2}$$

Null required scores fail this gate. After rejection, feedback d_t contains the actual probabilities, pass/fail verdicts, thresholds, and local errors:

$$H_{t+1} = H_t \parallel [(c_t, d_t)]. \tag{3}$$

The generator receives all records, oldest first and marked as not executed, before the unchanged original request. The decision model receives only the current request, schemas, and candidate. Readable schema-invalid candidates still receive binary checks; unreadable candidates receive local errors only. Reference answers never enter either model's requests.

A repair replaces the initial answer only after acceptance; if the limit is exhausted, DMFC returns c_0. No external function executes during this loop. Acceptance is a verifier decision, not a guarantee of benchmark correctness.

Algorithm 1 DMFC: verification and cumulative repair

Require: Request x, functions $\mathcal{F}$, generator G, decision model J
$H \leftarrow [\,]$
for $t = 0, \ldots, r$ **do**
  $c_t \leftarrow G(\text{Prefix}(H, x), \mathcal{F})$
  $(s, e, \text{readable}) \leftarrow \text{SchemaCheck}(c_t, \mathcal{F})$
  $p_f, p_a \leftarrow \text{null}, \text{null}$
  if readable **then**
    $p_f, p_a \leftarrow J(x, \mathcal{F}, c_t, \text{ApplicableChecks}(c_t))$
  end if
  if $\text{Accept}(s, c_t, p_f, p_a) = 1$ **then return** c_t
  if $t < r$ **then** $H \leftarrow H \parallel [(c_t, \text{Feedback}(e, p_f, p_a))]$
end for
return c_0

3 Experiment Setting & Results

3.1 Evaluation and Implementation

We evaluate 3,051 BFCL examples: simple Python (400), multiple (200), irrelevance (240), live simple (258), live multiple (1,053), live irrelevance (884), and live relevance (16). The AST checker compares names and arguments with allowed answers, not execution success. Irrelevance requires no call. Live relevance scores call presence only and is excluded from the 3,035-example main accuracy. Multiple means selecting among available functions; parallel, multi-turn, web-search, memory, and execution tasks are excluded.

Generators are `gpt-5-nano` (2025-08-07), `gpt-5.4-mini` (2026-03-17), and `claude-sonnet-5-5`. GPT uses reasoning effort `none`; Claude receives no thinking override. All use a 4,096-token output cap and automatic, nonparallel native tool choice. The decision model is `jev-1.13.0`; $r = 3$ and $\tau_f = \tau_a = 0.5$. These settings were chosen on the evaluated examples after thresholds 0.8/0.7 and last-candidate fallback; there is no held-out tuning set. Filtered or truncated generations count as incorrect.

For transfer, we adapt API-Bank [3] to first-call prediction: retain dialogue prefixes before their first API turn and supply a fixed 53-API catalog, independent of gold calls. Eight gold/schema conflicts are excluded before inference, leaving 204 cases grouped into development (48) and test (156). A common user instruction requests a native call candidate without execution. We score API names and type-normalized arguments; this exact-match adaptation differs from API-Bank's native execution score. The BFCL settings transfer unchanged. Four development cases verify the interface, without tuning thresholds or fallback; the prompt was revised after inspecting earlier test outcomes, so this evaluation is exploratory.

Compared methods. All methods share initial candidates. *Direct* returns the initial call; *Schema repair* uses local errors; *Self-refine* uses the generator's generic self-review for all r repairs, an adaptation rather than a reproduction of Self-Refine [6]. **DMFC-F** uses only function feedback and its acceptance gate; **DMFC-A** uses only argument feedback and its gate. Both compute applicable judgments but expose and require only their designated judgment, plus schema validation; DMFC-A accepts no call on schema alone. **DMFC-L** retains only the latest rejected candidate and feedback. Full **DMFC** retains all records. F/A isolate the two checks; L isolates history retention with the same thresholds, budget, and initial calls.

3.2 Main Experiment

Table 1 shows BFCL gains of 10.7 points for nano (paired 95% interval 9.7–11.6) and 1.0 for mini (0.5–1.5). Nano's largest category gain is on Irrelevance, 27.2% to 78.2%, driven by withholding unnecessary calls. Claude instead loses 1.3 points (interval −1.8 to −0.7): across three runs, repair fixes 58 initial errors but introduces 174 after accepted revisions. Its mean API time rises from 2.55 to 4.16 s.

Benchmark	Generator	Direct (%)	**DMFC** (%)	API s (D / M)	USD/1k (D / M)
BFCL	nano	64.1 ± 0.3	**74.8** ± 0.3	1.48 / 2.81	0.056 / 0.210
BFCL	mini	81.1 ± 0.2	**82.1** ± 0.1	1.30 / 1.98	0.703 / 1.137
BFCL	Sonnet 5.5	**88.3** ± 0.3	87.0 ± 0.3	2.55 / 4.16	4.168 / 6.665
API-Bank†	nano	60.0 ± 2.4	**66.5** ± 1.9	2.38 / 4.46	0.043 / 0.783
API-Bank†	mini	78.2 ± 0.6	**79.5** ± 0.6	1.49 / 2.03	0.814 / 1.460
API-Bank†	Sonnet 5.5	86.3 ± 0.4	**87.2** ± 0.0	1.81 / 2.79	20.827 / 24.580

Table 1: Direct/DMFC accuracy: mean ± sample s.d. of three runs, $r = 3$; bold compares methods per row. API seconds and USD/1k are Direct/DMFC pairs. BFCL: 3,035 accuracy cases, 3,051 time/cost cases; †API-Bank: 156 cases, exploratory adapted exact match. Time includes generation/verification only; cost omits 7 records with unavailable usage/prices. — means unavailable.

Method	Call	Func.	Args.	API s	USD/1k
Direct	63.8	74.8	81.2	1.30	0.056
Schema repair	65.8	76.4	81.8	1.36	0.059
Self-refine	70.0	80.3	82.2	5.47	0.265
DMFC-F	73.2	84.5	81.4	2.24	0.172
DMFC-A	72.5	82.6	**82.7**	2.21	0.169
DMFC-L	73.5	84.3	82.2	2.60	0.203
DMFC	**74.6**	**85.3**	82.1	2.61	0.211

Table 2: Ablation: one `gpt-5-nano` run, $r = 3$. Call/function accuracy (%) excludes live relevance. Arguments are scored only when the function is correct; bold marks each accuracy column's maximum.

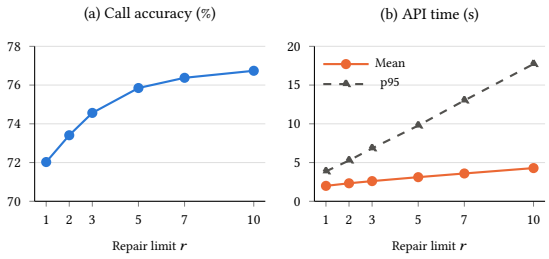


Figure 2: DMFC with `gpt-5-nano`: accuracy and API time over 3,051 examples from one run continued to $r = 10$ and cut at each limit. At $r = 3$, values match Table 2. API time sums generation and verification durations, excluding retry waits and tool execution.

API-Bank gains are 6.4 points for nano (interval 3.8–9.4), 1.3 for mini (0.2–2.6), and 0.9 for Claude (0.0–2.6). Claude's interval includes zero. These mixed effects do not establish a uniform benefit across generators or end-to-end tool-use success.

3.3 Ablation and Latency

In Table 2, both checks outperform either alone on call accuracy. Full history beats latest-only feedback by 1.1 points (95% interval 0.4–1.7), with similar mean API time (2.61 vs. 2.60 s). DMFC-A's higher conditional argument score does not imply better overall calls because the scored subsets differ.

Figure 2 cuts one continued trajectory at each repair limit; these are not independent runs. Increasing r from 3 to 10 raises accuracy from 74.6% to 76.7%, mean API time from 2.61 to 4.29 s, and p95 from 6.83 to 17.72 s; cost grows from \$0.211 to \$0.399 per 1,000 requests. The smaller accuracy gains therefore come with a larger latency tail.

Fallback and measurement limits. API time sums generation and verification, excluding retry waits, scoring, and tool execution. Cost includes initial and repair generation plus verification; unknown usage is omitted from known-cost means. Failed unsaved repair work is excluded, so costs are not invoice totals. Intervals bootstrap example IDs 2,000 times (seed 1729), averaging repetitions per ID. BFCL settings and API-Bank's input amendment followed inspected outcomes. API-Bank has only 156 test cases and no validated no-call negatives. Three generators, one verifier, single-run ablations, English-dominated public data, and ten retained BFCL annotation errors limit generalization. Native schema rejections count as incorrect. Claude BFCL concurrency changed from 8 to 32 with prefixes preserved; API times do not establish provider speed rankings. Acceptance and correctness remain separate.

4 Conclusion

DMFC combines function and argument judgments with cumulative rejected-call feedback before execution. DMFC improves the GPT comparisons but reduces Claude accuracy on BFCL, while adding API time and cost. The exploratory API-Bank first-call evaluation extends the evidence beyond BFCL but does not measure execution success. Independent evaluation, other decision-model families, and open-weight generators remain necessary.

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